

This is the configuration of needed files

- ✓ > pkdgrav3
- ✓ > sod_tube
- ✓ \$ cmakebuild.sh
- ✓ \$ cmakeshort.sh
- ≡ code
- ✓ \$ draw_compare_gif.sh
- \$ exportenv.sh
- ✓ \$ gif.sh

This is the folder from the bitbucket, only part of the subdirectories are displayed in this slide

```

└─ pkdgrav3
   ├── _deps
   ├── .venv
   ├── analysis
   ├── ANEOSmaterial
   ├── blitz
   └─ build

```

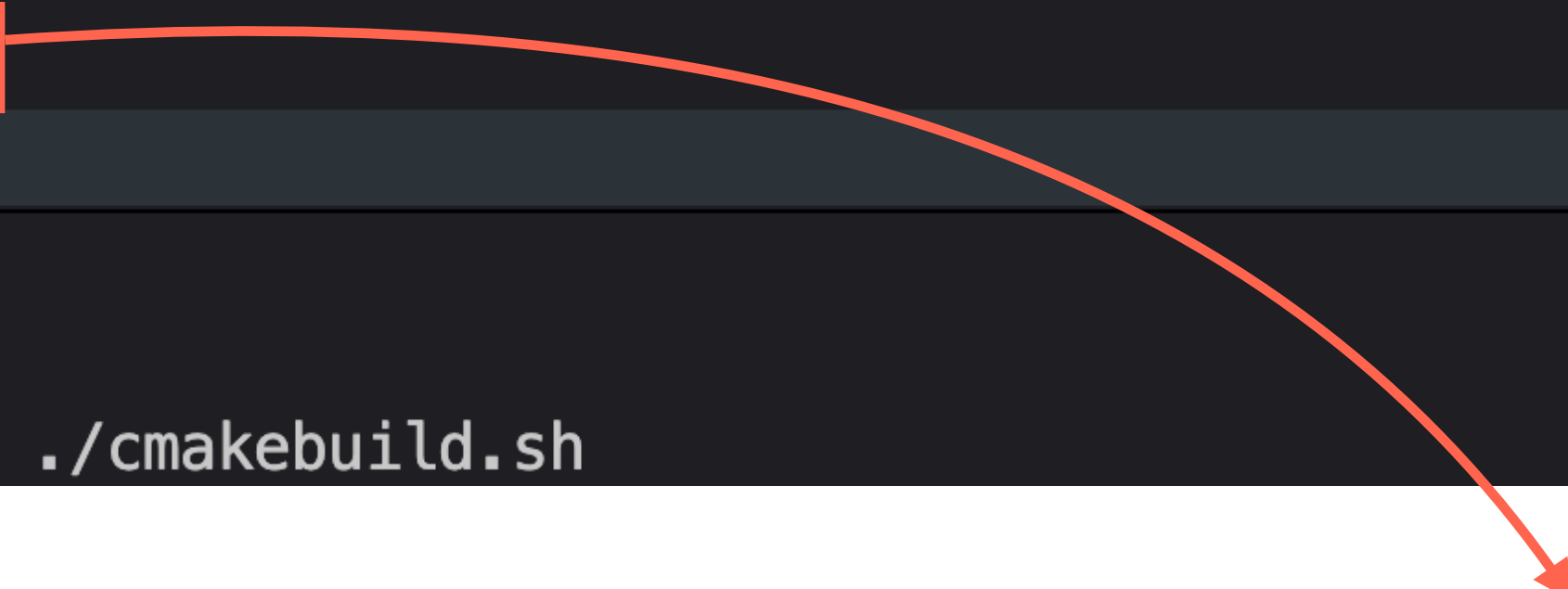
The executable file is build/pkdgrav3

```
grav2.cxx 3, M X
pkdgrav3 > gravity > grav2.cxx > queueCooling(PKD, workParticle *, bool)
210 static void queueCooling(PKD pkd, workParticle* wp, bool bGPU = true) {
211     //bGPU=false;
```

Uncommenting this line turns off GPU acceleration.

To generate a new test executable file, run `./cmakebuild.sh`

```
$ cmakebuild.sh ×  
$ cmakebuild.sh  
1  #!/bin/bash -l  
2  set -euo pipefail  
3  
4  uenv run --view=modules prgenv-gnu -- bash -lc '  
5      module load boost cmake cray-mpich fftw gcc gsl hdf5 python cuda  
6  
7      cd ~/pkdgrav3  
8      cd buildcpu  
9  
9  
TERMINAL  
● binliu@daint-ln004:~> ./cmakebuild.sh
```



This is the directory inside
which the new executable file lies

To update an already-made executable file and test it, run `./cmakeshort.sh`

```
$ cmakeshort.sh X
```

```
$ cmakeshort.sh
```

```
1  #!/bin/bash -l
2  set -euo pipefail
3
4  uenv run --view=modules prgenv-gnu -- bash -lc '
5      module load boost cmake cray-mpich fftw gcc gsl hdf5 python cuda
6
7      cd ~/pkdgrav3
8      cd buildcpu
9
10     python -m venv ../.venv
11     source ../.venv/bin/activate
12
13     python -m pip install -r ../requirements.txt
14
15     make -j 40
16 '
17 sbatch /users/binliu/sod_tube/peq_2_cpu.sh
```

This is the folder inside which
the new executable file lies

Other job scripts are also in `sod_tube/`

TERMINAL

```
● binliu@daint-ln004:~> ./cmakeshort.sh
```

```
$ peq_2_cpu.sh ×
```

```
sod_tube > $ peq_2_cpu.sh
```

```
24
```

```
25 cd /users/binliu/sod_tube
```

```
26
```

```
27 PKDGRAV3=/users/binliu/pkdgrav3/buildcpu/pkdgrav3
```

```
28
```

```
29 export OMP_NUM_THREADS=${SLURM_CPUS_PER_TASK}
```

```
30 export MPICH_GPU_SUPPORT_ENABLED=1
```

```
31
```

```
32 srun --cpus-per-task=${SLURM_CPUS_PER_TASK} \
```

```
33 | "${PKDGRAV3}" ./initial/peq_2.par
```

This is the executable file

Other configuration files are also in sod_tube/initial/

The relevant parameters in .par files are

```
✓ achOutName = "peq_kwheat"

# Memory and performance
bMemUnordered = 0 # iOrder replaced by potential and group id
bNewKDK       = 1 # No accelerations in the particle, dual tree possible
bDualTree     = 0 # Use two trees
bStandard     = 0
bMemSoft      = 1 # include the softening in the particles

# Initial Condition
dBoxSize      = 1000      # Mpc/h
# nGrid       = 128      # Simulation has nGrid^3 particles
# b2LPT       = 1        # second order IC
bWriteIC      = 1        # Write ICs to file
iSeed         = 314159265 # Random seed

dPeriod       = [1.0, 0.5/128*16, 0.5/128*16]

✓ achInFile    = "./initial/peq_ic_grid_isentrope_relaxed1600_combined.tipsy"
✓ achOutPath   = "./results/" + achOutName + "/{name}.{step:05d}{type}"
```

Other snapshots of
initial particles are
also in `sod_tube/initial/`

✓ # Logging/Output
iLogInterval = 1
iOutInterval = 1
#iCheckInterval = 10
bDoDensity = 0
bVDetails = 1
dHubble0 = math.sqrt(math.pi*8/3)

bOverwrite = 1
bParaRead = 1 # Read in parallel
bParaWrite = 0 # Write in parallel (does not work on all file systems)
#nParaRead = 8 # Limit number of simultaneous readers to this
#nParaWrite = 8 # Limit number of simultaneous writers to this

Cosmological Simulation

bComove = 0 # Use comoving coordinates
bPeriodic = 1 # with a periodic box
bEwald = 0 # enable Ewald periodic boundaries
bDoGravity = 0

Accuracy Parameters

bEpsAccStep = 1 # Choose eps/a timestep criteria
dTheta = 0.40 # accuracy of forces for $z > 20$
dTheta20 = 0.55 # accuracy of forces for $z > 2$
dTheta2 = 0.70 # accuracy of forces for $z \leq 2$
dTheta = 0.7

parameters for hydrodynamic solvers

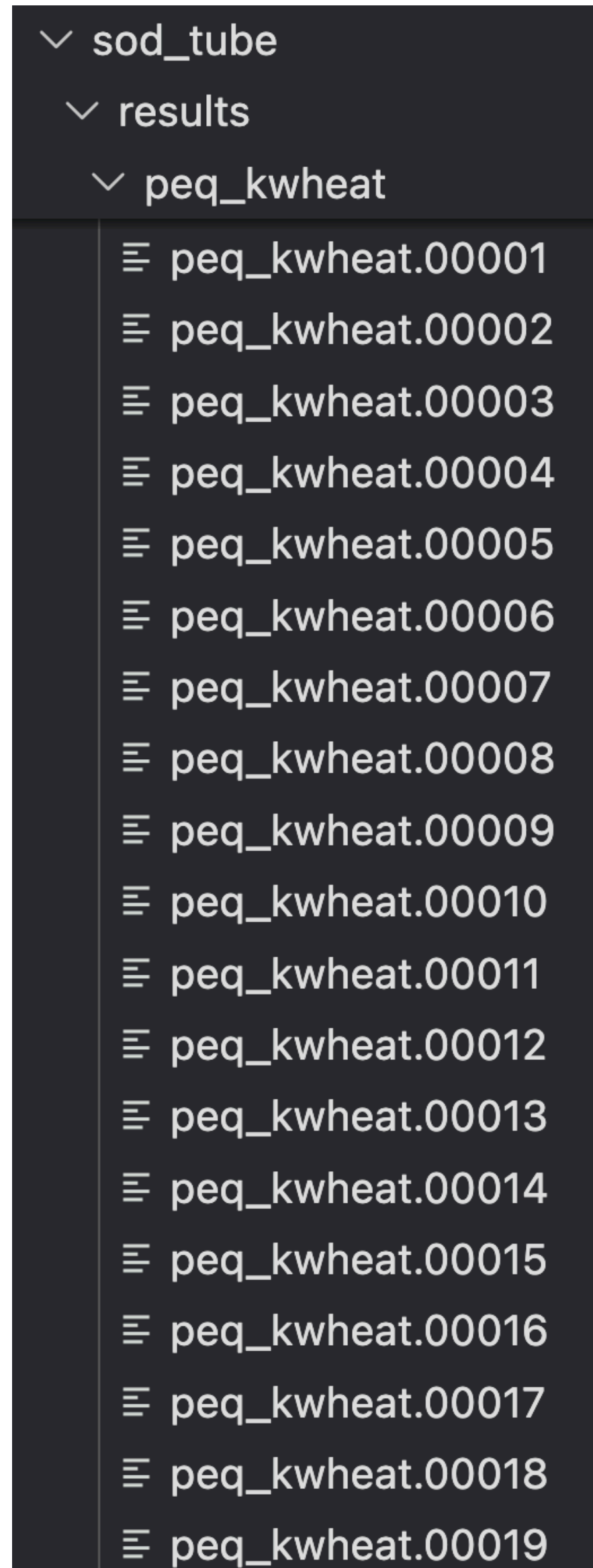
hydro_model: 0=NONE, 1=SPH, 2=MFM, 3=MFV

hydro_model = 1
bMeshlessHydro = 0 # legacy meshless switch (kept for compatibility)

cooling parameters

✓ cooling_model = 2 # 0: no cooling, 1: KWH cooling, 2: KWH cooling with heating,

The default folder of output snapshots are in sod_tube/results/



Only part of the snapshots are listed here

./gif.sh can generate gif of density, pressure, and temperature evolution of a given simulation

```
$ gif.sh
$ gif.sh
1  #!/bin/bash -l
2
3  set -e
4
5  cd /users/binliu/sod_tube
6
7  uenv run --view=modules pkdgrav3/3.4 -- bash -lc '
8      module load gcc pkdgrav3-python
9      source .venv-plot/bin/activate
10     export MPLBACKEND=Agg
11     python plot_gif.py ./results peq_kwheat 1 100 10
12 '
```

```
▼ sod_tube
  ▼ results
    > peq_kwh
    ▼ peq_kwheat
      🖼️ peq_kwheat_compare_density.gif
      🖼️ peq_kwheat_compare_pressure.gif
      🖼️ peq_kwheat_compare_temperature.gif
```

The second parameter is the last snapshot number (starting from zero)

The last parameter is the temperature

The first parameter is the stride of snapshots between two frames

`./draw_compare_gif.sh` generate gif of density, pressure, and temperature, where `cooling_model=0,1,2` are on the same plot.

```
● binliu@daint-ln004:~> ./draw_compare_gif.sh peq 1 10 14400
```

The meaning of parameters is the same as `gif.sh`

